Audio–EMG Classification of Instructed Jaw Activities Relevant to Tooth-Contact Bruxism with a Dual-Branch Wavelet CNN

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Abstract

Current consensus defines bruxism as masticatory-muscle activity that may include clenching or grinding with tooth contact and bracing or thrusting without tooth contact. A device that classifies instructed tooth-contact tasks therefore does not, by itself, diagnose bruxism. Here, we report a proof-of-concept engineering study of whether surface electromyography (EMG), audio, and a dual-branch wavelet convolutional neural network can distinguish quiet rest from four instructed jaw-activity classes. Five adults with a prior clinical diagnosis of bruxism completed awake laboratory recordings of rest, jaw movement, clenching, grinding, and chewing. The activities were deliberately performed rather than observed spontaneously. A signal-quality audit showed that the acquisition hardware had already removed the 60-Hz mains fundamental and that 91.3–99.8% of the in-band power of the previously filtered EMG was residual mains *harmonics*; the analysis was therefore repeated with a harmonic notch bank, trigger-constrained window labeling, and leakage-free nested leave-one-subject-out (LOSO) evaluation. Averaged over the five held-out participants and three training seeds, the fused five-class model achieved 82.1% accuracy, 73.6% macro-F1, and a macro one-vs-rest ROC–AUC of 0.962. A sensitivity analysis excluding chewing yielded 79.8% accuracy and 77.9% macro-F1 across rest, movement, clenching, and grinding. Collapsing the predictions post hoc into instructed clenching/grinding versus rest/movement/chewing yielded 93.6% accuracy, an F1 of 87.3%, and a ROC–AUC of 0.966, with 6.0% of quiet-rest windows falling into the clenching/grinding group. A modality ablation, in which only the modality varied and the unused branch was removed, placed EMG alone at 72.0% macro-F1 and audio alone at 42.8%, and adding audio to EMG changed macro-F1 by less than the spread between training seeds. A subsequent audit of the microphone channel itself, however, found that it is not per-participant audio: the 100 recordings contain only 37 distinct microphone waveforms, 83 of them bit-identical after a circular rotation to another participant's recording of the same condition, and the channel is unaligned with the EMG and confined below

10 Hz. The ablation numbers are reported as run and bound what such a channel contributes; they do not measure what a microphone contributes, and RQ2 is recorded as unanswered. The EMG channels are distinct across every recording and no other result in this paper reads the microphone. The baseline architecture comparison has not been run on the corrected signal and is reported as pending rather than estimated from the superseded pipeline. Including quiet rest allows activity windows to be scored against a non-event condition rather than only against one another. However, the rest data were recorded in a controlled laboratory and do not represent the full range of unrestricted background behavior. No spontaneous awake- or sleep-bruxism episodes were recorded. These results establish only the technical feasibility of multimodal classification of instructed jaw activities and quiet rest in this five-participant dataset; they do not establish continuous free-living event detection, population-level generalization, or clinical validity for diagnosing awake or sleep bruxism.

Keywords: Jaw-activity classification, Tooth contact, Surface electromyography, Audio sensing, Convolutional neural networks, Wavelet decomposition, Sensor fusion, Powerline interference, Leave-one-subject-out validation

1 Introduction

Bruxism terminology has changed substantially. International consensus no longer defines bruxism simply as involuntary tooth grinding or clenching. Bruxism is a repetitive masticatory-muscle activity characterized by clenching or grinding of the teeth and/or by bracing or thrusting of the mandible [1, 24]. Sleep bruxism and awake bruxism are distinct circadian manifestations. Sleep bruxism consists of rhythmic (phasic) or non-rhythmic (tonic) masticatory-muscle activity during sleep, whereas awake bruxism consists of repetitive or sustained tooth contact and/or mandibular bracing or thrusting during wakefulness [24]. Tooth contact is therefore neither necessary for all bruxism activities nor sufficient to establish that an observed activity is bruxism.

Current frameworks also treat bruxism as a behavior whose clinical relevance depends on the individual and context, rather than as a disorder that is uniformly pathological. Bruxism activities may be neutral, may act as risk indicators for adverse oral-health outcomes, or may have protective associations in some circumstances [1, 24]. A recent biopsychosocial model emphasizes that biological, psychological, and social influences interact across the 24-hour cycle and that motor activity cannot be interpreted apart from the person's broader clinical context [25]. This wider perspective has led to the description of bruxism as a medical gateway for dentistry: careful assessment may connect oral findings with sleep, pain, psychological, medication-related, and other health factors, rather than reducing the problem to dental mechanics alone [26].

Assessment must therefore be multidimensional. The Standardised Tool for the Assessment of Bruxism (STAB) combines subject-based report, clinical examination, instrumental assessment, possible consequences, and risk or associated factors [2]. Instrumental sensors can contribute objective information about muscle activity, but they represent only one component of this framework. This distinction is particularly important for temporomandibular disorders (TMDs). Bruxism and TMD are not interchangeable diagnoses, and the present study was not designed to evaluate TMD. The INfORM/IADR standard-of-care statement describes TMDs as heterogeneous, multifactorial biopsychosocial conditions whose diagnosis requires validated history and clinical assessment and whose management should generally be patient-centered, conservative, and reversible [27]. Accordingly, the presence of a jaw-muscle pattern alone should not be presented as a diagnosis of TMD or as proof that bruxism caused a patient's pain or dysfunction.

Recent surveys and educational reports show why precise terminology matters. Dental practitioners use heterogeneous bruxism assessment and management strategies, and outdated beliefs remain common even among trainees [28, 29]. Similar educational gaps have been documented for TMDs [30]. Engineering studies can help address an important measurement problem, but only if the output of a classifier is separated clearly from a clinical diagnosis.

Instrumental assessment outside the clinic remains technically challenging. Surface EMG is sensitive to masticatory-muscle activation but can respond similarly to functional and non-functional activities. Audio may add information about tooth contact and mastication, but it is also sensitive to ambient noise and raises privacy considerations. Natural behaviors fluctuate within and across days; ecological momentary assessment (EMA) studies have therefore used repeated sampling in the natural environment rather than relying on a single retrospective report [32, 33]. Proposed research routes for awake-bruxism metrics combine subject-based, clinically based, EMA, and instrumental approaches and emphasize matching the measurement method to the research question [31]. Multi-day, 24-hour EMG recording has also been proposed for characterizing muscle activity in the home environment [3].

Against this background, the present study addresses a narrow engineering question. It does not evaluate spontaneous awake bruxism, sleep bruxism, disease severity, TMD, or treatment response. Instead, it tests whether an audio–EMG system can distinguish quiet rest and four instructed jaw activities performed in a controlled laboratory by five adults with a prior clinical diagnosis of bruxism. The activity labels describe the experimental conditions, not independently adjudicated bruxism episodes.

A second, methodological point motivates this study. Instrumental studies of masticatory-muscle activity are reported almost exclusively through summary accuracies, and the measurement chain that produced them is rarely auditable. During the preparation of this analysis, a spectral audit of our own recordings showed that the conventional preprocessing recipe applied to them—notch the mains fundamental, then band-pass—had removed the one frequency the acquisition hardware had already suppressed and had left the mains *harmonics*, which carried the overwhelming majority of the in-band power, entirely intact. We report that finding, its correction, and its effect on the results, because a classifier trained through such a channel can appear to work while separating interference rather than physiology.

The study addresses three research questions:

- **RQ1:** Can off-the-shelf surface EMG and microphone sensors provide signals that support five-class classification of quiet rest, jaw movement, clenching, grinding, and chewing?
- **RQ2:** Within this dataset, how much does audio add to EMG, and is the improvement concentrated in the chewing class?
- **RQ3:** Does a dual-branch CNN with modality-specific wavelet decomposition outperform the evaluated single-pathway and conventional baselines under the same leave-one-subject-out protocol, including when the task is collapsed to instructed clenching/grinding versus the three comparison conditions?

RQ1 is answered below; RQ2 and RQ3 are *not*. All three were previously estimated from runs that used the defective filter chain, and those estimates are withdrawn rather than restated. The confirmatory modality ablation for RQ2 was subsequently run on the corrected signal and is reported here, but a later audit of the microphone channel found it to be duplicated across participants, unaligned with the electromyogram and confined below 10 Hz (Section 4.2); the ablation is therefore reported as run and bounds what such a channel contributes, while RQ2 itself is recorded as unanswered by this dataset. The baseline architecture sweep for RQ3 is prespecified but has not been run, and where an interim indication exists it comes from a declared screening harness and is labeled as such throughout. Nothing in either case affects the electromyographic results: the EMG channels are distinct across all 100 recordings and every result outside the modality ablation reads EMG alone.

2 Prior technical advances

2.1 EMG-based assessment

Surface EMG is widely used for ambulatory assessment because it measures masticatory-muscle activation directly [6]. Early portable systems established the feasibility of home recording but had limited specificity for distinguishing target activity from other orofacial activity [5]. Castroflorio et al. [7] combined EMG and electrocardiographic information to identify sleep-bruxism episodes in a natural environment.

Machine-learning studies have reported high performance on controlled tasks, although direct comparison is difficult when labels, participants, and validation protocols differ. Sonmezocak and Kurt [8], for example, reported greater than 90% accuracy using EMG features across jaw opening, clenching, rhythmic grinding, and fatigue/pain classes. Gul et al. [4] evaluated sleep-related recordings across postures, while Heyat et al. [9] combined several physiological modalities. Lan et al. [10] described different surface-EMG patterns during centric and eccentric instructed activities. These studies support technical feasibility but do not make their reported accuracies interchangeable, because some classify pre-segmented tasks whereas others aim to identify events within continuous recordings.

2.2 Acoustic and wearable sensing

Acoustic sensing may complement EMG when oral activities produce different sound patterns. Romano et al. [11] showed that signals from voluntary oral behaviors depend strongly on transducer location and identified ear-adjacent placement as promising for longer recordings. Martinot et al. [12] used mandibular-movement analysis to identify phasic sleep-bruxism events in patients with obstructive sleep apnea. Patented systems have also used adaptive sound thresholds or chin-mounted accelerometers [13, 14]. For sleep applications, however, portable devices require comparison with an appropriate sleep reference method; a controlled awake-task classifier cannot be assumed to transfer to sleep [34].

2.3 Signal processing and deep learning

Wavelet decomposition is well suited to non-stationary EMG because it preserves joint time–frequency information [15]. Wavelet features have performed well in EMG classification [16], and filter design remains important for preserving physiological content while suppressing noise [17]. CNNs can learn local patterns in physiological signals [18, 19]. Focal loss and class-weighting methods can reduce the influence of class imbalance [20, 21]. The remaining engineering gap addressed here is whether modality-specific wavelet branches improve classification of quiet rest and the four instructed activities in the present dataset.

Surface-EMG reviews consistently identify powerline interference as the dominant environmental artifact and recommend notch filtering combined with a band-pass stage [6, 17]. That recommendation is usually implemented as a single notch at the supply frequency. It is only valid when the supply fundamental is in fact the dominant interference. Head-mounted electrodes with long leads, in a laboratory containing displays, lighting and switched-mode supplies, can instead pick up a harmonic-rich waveform, and some acquisition systems apply their own fixed notch before the data reach the analysis software. In that combination the textbook chain removes nothing and still looks correct on a filter-response plot. Verifying the filter against the measured spectrum of the actual recordings—rather than against its own design—is therefore a necessary step, and it is the step added here.

3 Methods

This proof-of-concept study was approved by the Institutional Review Board of the University of Rhode Island (Protocol #IRB22275690-2). All methods were performed in accordance with relevant institutional guidelines and regulations. Written informed consent was obtained from every participant, including consent to publish the identifying image in Fig. 1.

3.1 Participants and study scope

Five adults (2 women and 3 men; age 38.0 ± 10.4 years) were recruited through flyers, messages to local oral-care and physical-therapy providers, and word of mouth. Eligibility required a prior clinical diagnosis of bruxism. The study did not independently re-diagnose or phenotype that history, and the available study records did not stratify participants by sleep bruxism, awake bruxism, grinding, bracing, or thrusting predominance. The prior diagnosis therefore describes the sample but is not a reference standard for the activity labels used in this analysis.

All analyzed tasks were performed intentionally while participants were awake in the laboratory. The study did not record spontaneous awake-bruxism episodes or sleep. Consequently, the labels "clenching" and "grinding" below refer to instructed task performance. They should not be interpreted as natural or independently verified bruxism episodes.

3.2 Experimental procedure

The session began with written consent and a pre-screening survey. Surface EMG electrodes, an omnidirectional microphone, and a camera were positioned and secured. Participants first completed 1 minute of quiet rest. They then performed five repetitions of jaw opening/closing, left/right deviation, protrusion/retrusion, and left/right biting. This was followed by 1-minute trials of isometric molar and incisor clenching, alternating 3 seconds of activity with 3 seconds of relaxation.

After a break, participants completed 3-minute trials of instructed grinding, chewing cheese, chewing baby carrots or popcorn, and chewing gum. Each task was repeated three times, with a 1-minute rest between repetitions. A handheld button transmitted event markers synchronized with the physiological recordings. Participants pressed the button at task initiation, and the experimenter visually confirmed task onset.

3.3 Window labeling and analyzable sample

The button and task protocol established experimental task labels, not clinical bruxism labels. Every analyzed window is required to lie *wholly inside* one interval during which the trigger marks the participant as actively performing the named task, so that no window can span two conditions or a trigger transition. Windows are cut from the continuously filtered recording, never filtered individually, because filtering each window separately injects an edge transient exceeding one signal standard deviation into the first 50 samples of every example.

Three exclusions are applied before windowing. A **transition guard** of 0.25 s is removed on each side of every trigger edge. The choice was measured rather than assumed: over 432 task onsets (358 with a detectable envelope transition) the median lag between the button press and the electromyographic onset was -0.075 s, muscle activity preceded the marker at 79.9% of onsets, and among the 20% of onsets in the harmful direction the median lag was 0.054 s and the largest ever measured was 0.352 s. A **startup guard** of 0.5 s is removed at the beginning of every recording, replacing an earlier unexplained 3-s skip; amplifier settling transients were observed in 66 of 100 recordings, all settled within 0.40 s, with peak excursions up to 12,580$\times$ the recording's robust scale. Finally, quiet-rest windows are drawn only from the five dedicated rest recordings; trigger-off intervals inside active recordings are *not* treated as rest, because they are unlabeled rather than verified as quiet.

The resulting rest class represents quiet laboratory rest only. It does not sample the full range of everyday non-event behaviors and therefore does not turn this window classifier into a validated continuous detector.

3.4 Sensors and dataset

Four differential surface-EMG channels were recorded from two bilateral bipolar electrode pairs, one pair per side of the head. The tentative electrode map records them as left masseter, left temporalis, right masseter, and right temporalis. One omnidirectional microphone was positioned near the temporomandibular joint (Fig. 1). All signals were sampled at 1200 Hz. The acquisition chain documented no physical calibration for either modality, so all amplitudes in this paper are reported in arbitrary analog-to-digital converter units and are deliberately not labeled µV, Pa, or dB.

The microphone column requires a further description, because what it contains is not what the sensor list implies. It is integer-valued with a quantization step of exactly 1 count and a range of roughly 50–227; 96% of its power lies below 10 Hz, and its discriminative content is concentrated at 1–3 Hz, the masticatory rhythm. It therefore behaves as a rectified sound-level or envelope output rather than as an acoustic waveform, and at a 1200-Hz sampling rate the Nyquist frequency of 600 Hz sits below the band in which tooth contact radiates most of its energy in any case. Section 4.2 reports the remaining properties of this channel, which are more consequential still. Transducer model, output type, preamplifier gain and converter resolution are not documented for either modality.

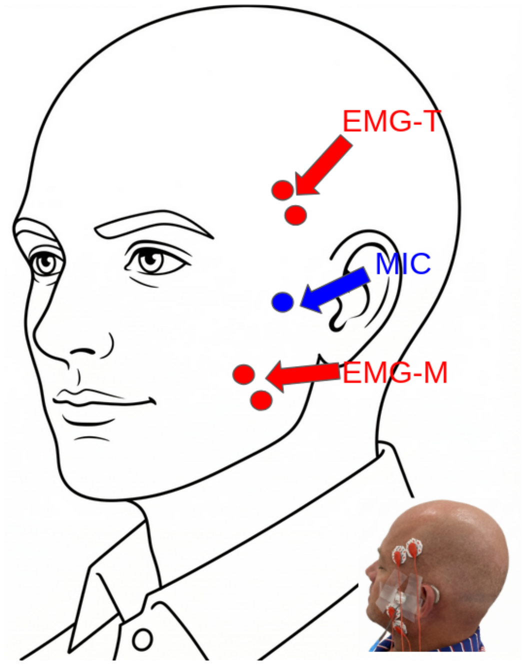

Fig. 1 Sensor placement for multimodal data acquisition. Bipolar surface-EMG electrode pairs were positioned over the temporalis (EMG-T) and masseter (EMG-M) muscles. A microphone (MIC) was secured near the temporomandibular joint. This arrangement was evaluated only during instructed awake laboratory tasks.

The recordings were consolidated into five experimental classes. **Rest** comprised windows from the five dedicated quiet-baseline recordings only; as described above, inter-trial intervals inside active recordings were not admitted. **Movement** included opening/closing, lateral deviation, and protrusion/retrusion. **Clenching** included molar clench, incisor clench, and unilateral left/right bite. **Grinding** included the instructed grinding trials. **Chewing** included cheese, carrots or popcorn, and gum. These categories were selected to contrast quiet non-event periods, static loading, lateral tooth-contact activity, non-chewing jaw movement, and functional mastication. They were not intended as a comprehensive taxonomy of bruxism. In particular, mandibular bracing and thrusting without tooth contact were not elicited as distinct classes.

Chewing was included because it is a common functional activity and a plausible confound for an ambulatory jaw-activity classifier. Because it may also be comparatively easy to recognize, the primary five-class results are accompanied by a sensitivity analysis that excludes chewing. A second analysis collapses the five labels into instructed clenching/grinding versus rest/movement/chewing.

Continuous signals were segmented into 1.0-second windows (1200 samples) with a 0.5-second stride (600 samples), creating 50% overlap between adjacent windows. Under the containment and guard rules above, 100 recordings yielded 6,173 analyzable windows from 99 recordings: 590 rest, 333 movement, 799 clenching, 816 instructed grinding, and 3,635 chewing (Table 1, Fig. 2). These counts supersede the 11,845 windows reported previously, which were produced by a legacy policy that labeled whole recordings rather than trigger-verified intervals and therefore included inter-trial and transition material inside activity classes. The two sets of counts are not comparable, and only the trigger-constrained counts are used here.

These windows are not independent observations: adjacent windows overlap by 50%, many windows originate from the same recording, and all of them originate from only five participants. The participant, rather than the window, therefore defines the effective unit for generalization, and all primary metrics below are computed within a participant before being summarized.

Table 1 Dataset characteristics under the trigger-constrained labeling policy.

Property	Value
Participants	5
Recordings (contributing)	100 (99)
EMG channels	4 (2 bilateral pairs)
Microphone channels	1
Sampling rate	1200 Hz
Signal units	Arbitrary ADC units
Window size	1200 samples (1.0 s)
Stride	600 samples (0.5 s)
Transition guard	0.25 s each side
Startup guard	0.5 s per recording
Rest windows	590
Movement windows	333
Clenching windows	799
Grinding windows	816
Chewing windows	3,635
Total analyzed windows	6,173
Experimental classes	5
Quiet-rest class	Included

3.5 Mains interference and the corrected filter chain

Every result in this paper was regenerated after a spectral audit of the raw recordings, which is reported here because it changed the analysis substantially.

The acquisition software applied its own fixed notch, recorded in every metadata sidecar, and that notch removed the 60-Hz fundamental before the data were written to disk. Interference survived at the harmonics instead. Measured against the local noise floor of the raw signals, the peaks at 180, 300, and 420 Hz ranged from tens of times that floor in the least affected participant to more than a million times it in the most affected. Across all 100 recordings, 62 had more than 30% of their raw in-band EMG power concentrated within ± 3 Hz of a mains multiple, and in the five dedicated quiet-rest recordings that share was 91.3–99.8%.

The preprocessing chain used in earlier analyses of this dataset was a single 60-Hz notch ($Q = 30$) followed by the 20–450-Hz band-pass. On the recordings described above it removed the only mains frequency that was already absent and passed every frequency that was present: the mains share of in-band power after filtering was 91.4–99.8%, essentially unchanged from the raw signal (Fig. 3a, c). A filter-response plot of that chain looks entirely conventional, which is why the problem was not visible until the response was drawn over the measured spectrum of the data.

The corrected chain notches every mains multiple inside the pass band—60, 120, 180, 240, 300, 360, and 420 Hz—before the same fourth-order Butterworth band-pass from 20 to 450 Hz. Each notch is *constant-width* (8 Hz at -3 dB, so $Q = f_0/8$) rather than constant-Q. Constant Q would give a 2-Hz notch at 60 Hz and a 14-Hz notch at 420 Hz, that is, the narrowest response exactly where the hardware notch left a residue and the widest where nothing survives. Both variants remove 56 Hz of the 430-Hz pass band, an identical 13% cost, but at that same cost the constant-width bank left the worst-contaminated cell at $3.2\times$ its local noise floor against $25.6\times$ for constant Q, an eight-fold reduction in residual interference. After correction, the mains share of in-band power in the quiet-rest recordings fell to 0.46–0.66% (Fig. 3b, c). A comb filter and spectral interpolation were also implemented and measured on the same windows; both left more residual interference, and spectral interpolation is additionally acausal by construction. The filter chain was chosen on this interference criterion, which can be specified in advance, and not on held-out classification score.

The microphone signal received a second-order 20-Hz high-pass to remove its large DC offset. No further spectral shaping was applied, because the transducer's frequency response is undocumented. The 5-Hz high-pass stage described in the earlier version of this pipeline followed a 20–450-Hz band-pass and was therefore a no-op; it was removed rather than carried forward. The cost of the surviving high-pass was measured only afterwards and is stated here rather than in the limitations: because 96% of this channel's power lies below 10 Hz, the stage retains a median of 1.19% of its variance (range 0.50–3.25%), and it removes the 1–3-Hz band that carries the channel's entire separation between conditions. What it keeps is at or below the 1-count quantization floor for the clenching and grinding conditions. Every reported run used this chain; it is described as defective rather than replaced, because replacing it would not repair the channel (Section 4.2) and would silently change published numbers.

All filtering is zero-phase (forward–backward second-order sections), applied to the whole continuous recording before any window is cut. Zero-phase filtering reads samples from the future of each output sample. It is appropriate for the offline analysis reported here and it cannot support a streaming, wearable, or real-time claim; a causal mode exists in the implementation and the mode used is recorded in every run bundle.

3.6 Normalization and wavelet decomposition

EMG and microphone signals were z-scored per channel using means and standard deviations computed *only* from the training participants of the fold in question. The list of participants that produced each fold's statistics is stored with the fold and asserted not to contain the held-out participant before that participant is evaluated. A per-participant (transductive) normalization scope is implemented and prespecified as a separate configuration, but it is not used for any

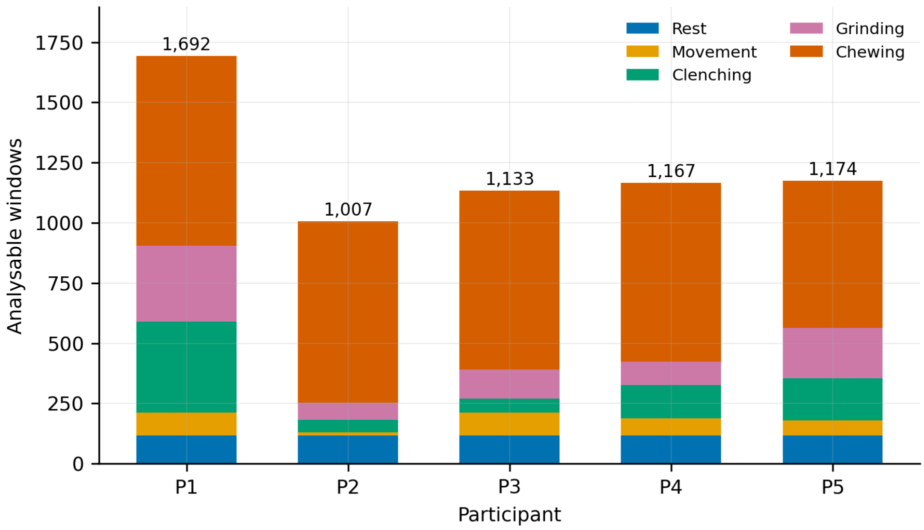

Fig. 2 Analyzable windows per participant and class after the containment, transition-guard and startup-guard exclusions. Participants are labeled P1–P5 here and in every later figure, in the same order as Table 7. Class sizes are strongly unequal both between classes—chewing supplies 58.9% of all windows—and between participants within a class, which is why class weighting, minority-only augmentation and participant-level aggregation are all used.

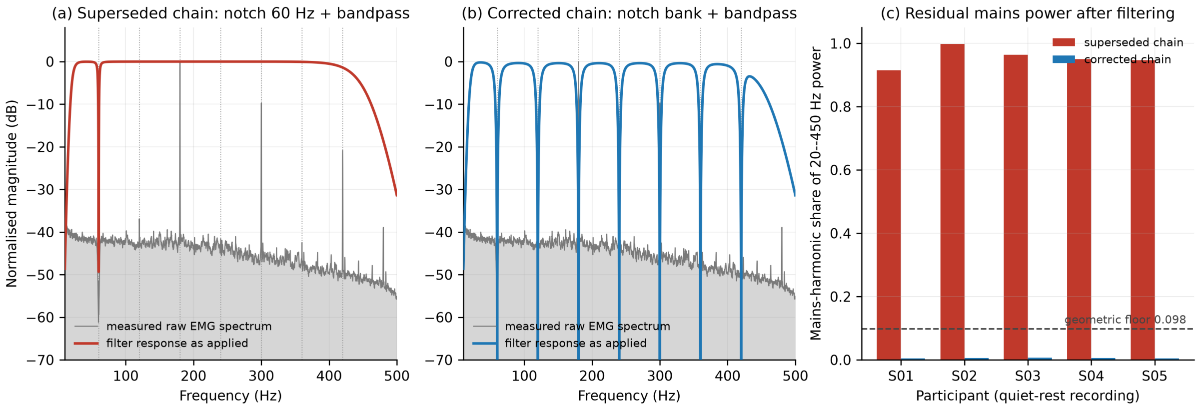

Fig. 3 The mains-harmonic defect and its correction. (a) The superseded chain: the 60-Hz notch lands where the acquisition hardware had already removed the fundamental, while the tall harmonic peaks at 180, 300 and 420 Hz pass through the band-pass untouched. (b) The corrected chain: seven constant-width notches land on the peaks. Gray traces are the mean raw EMG spectrum of the five dedicated quiet-rest recordings; colored traces are the zero-phase response as actually applied; dotted lines mark 60 Hz and its harmonics. (c) Share of 20–450-Hz power lying within ±3 Hz of a mains multiple, per participant, after each chain. The dashed line marks the 0.098 share a perfectly white signal would score, because seven ±3-Hz bands occupy 42 Hz of a 430-Hz pass band; a notch bank scores below it by deleting those bins.

number reported here; the strict scope is the one that would support a deployment claim without a fitting session.

Figure 4 shows the chain on a real excerpt. EMG signals were then decomposed using the Daubechies-4 wavelet at four levels, and the EMG sub-branches used A_4 (< 38 Hz), D_3 (75–150 Hz), and D_1 (300–600 Hz, of which only the 300–450-Hz portion survives the band-pass). Daubechies wavelets were selected because of their compact support and prior use in transient EMG analysis [15]. Audio signals were decomposed using the Coiflet-5 wavelet at five levels, with sub-branches A_5 (< 18.75 Hz), D_3 (75–150 Hz) and D_1 (300–600 Hz), selected in preliminary testing for its time–frequency representation of transient acoustic features [23]. The A_5 band lies entirely inside the stopband of the 20-Hz high-pass described above and retains 2.95% of its power, so that sub-branch receives the high-pass roll-off residue—baseline wander—which the batch normalization that follows it then rescales to unit variance. The contradiction was present in every reported run and was not noticed until the two stages were checked against each other rather than separately. It is retained here so that those runs reproduce, and the pipeline now refuses any band whose own filter chain has removed it unless the configuration declares the exception in writing.

3.7 Model architecture

The model processed EMG and audio in separate branches before fusion (Fig. 5). The EMG branch contained three CNN sub-branches, one per retained wavelet band. Each sub-branch applied a 3-tap convolution to 8 channels, batch normalization, a rectified linear unit, max pooling by 2, a second 3-tap convolution to 16 channels, batch normalization, a rectified linear unit, and adaptive average pooling to a single value per channel, contributing 16 features per band and 48 EMG features in total. The audio branch used an analogous structure with 4 and 8 channels and produced 24 features.

The 72 concatenated features were passed to a fusion multilayer perceptron (MLP) with dimensions $72 \rightarrow 48 \rightarrow 32$. Batch normalization, a rectified linear unit, and dropout with probability 0.5 were applied in each hidden layer. A final linear layer produced logits for rest, movement, clenching, grinding, and chewing. The complete model has 7,485 trainable parameters (1,656 in the EMG branch, 432 in the audio branch, 5,232 in the fusion MLP, and 165 in the classification head), or 29.2 KiB at 32-bit precision. In the modality ablations the unused branch is not constructed at all, so its parameters are genuinely removed rather than fed zeros: the EMG-only model has 5,901 parameters and the audio-only model 3,525. Adding the microphone pathway to the EMG-only model therefore costs 1,584 parameters, or 6.2 KiB.

3.8 Training and evaluation

To address class imbalance after adding rest, the model used inverse-frequency class weights recomputed within each outer training fold, and augmentation of minority-class training windows by amplitude scaling in $[0.9, 1.1]$ (applied with probability 0.5), additive Gaussian noise at 2% of the window standard deviation (probability 0.3), and circular shifts of up to ± 50 samples (probability 0.3). A window was eligible for augmentation with probability 0.4 and only if its class fell below 70% of the largest class in that fold, so chewing windows were never augmented and rest windows were. Augmentation was applied only within the training partition of each fold, enforced structurally: the augmenter raises an error for any stage other than training, and each transformation is seeded by the triple (run seed, epoch, sample identifier) so that it does not depend on worker count or batch order.

Optimization used AdamW with weight decay 1×10^{-4}, gradient clipping at norm 1.0, and batch size 64. The loss function and learning rate were selected inside each fold (below) from focal loss [20] with focusing parameter $\gamma = 1.5$ or cross-entropy, and from 3×10^{-4} or 1×10^{-3}. Table 2 lists these settings together with the prespecified configurations of the architecture comparison.

Evaluation used *nested* leave-one-subject-out (LOSO). In each of the five outer folds, one participant was sealed as the test set and the remaining four entered a participant-grouped inner LOSO loop of four inner folds. Every normalization statistic, class weight, augmentation minority set and hyperparameter was fitted on inner-training participants only. The best epoch was recorded per inner fold by validation macro-F1, and the epoch budget for the final refit was the median of those best epochs. The model was then refit on all four outer-training participants for exactly that many epochs, with no early stopping and no validation set, and the held-out participant was evaluated exactly once. This is 17 model fits per outer fold (4 trials $\times$ 4 inner folds, plus the refit). The held-out participant's sample identifiers are stored privately and released only by an explicit single-use call, so selection code cannot reach them even accidentally. Macro-F1 was the selection objective rather than accuracy, because

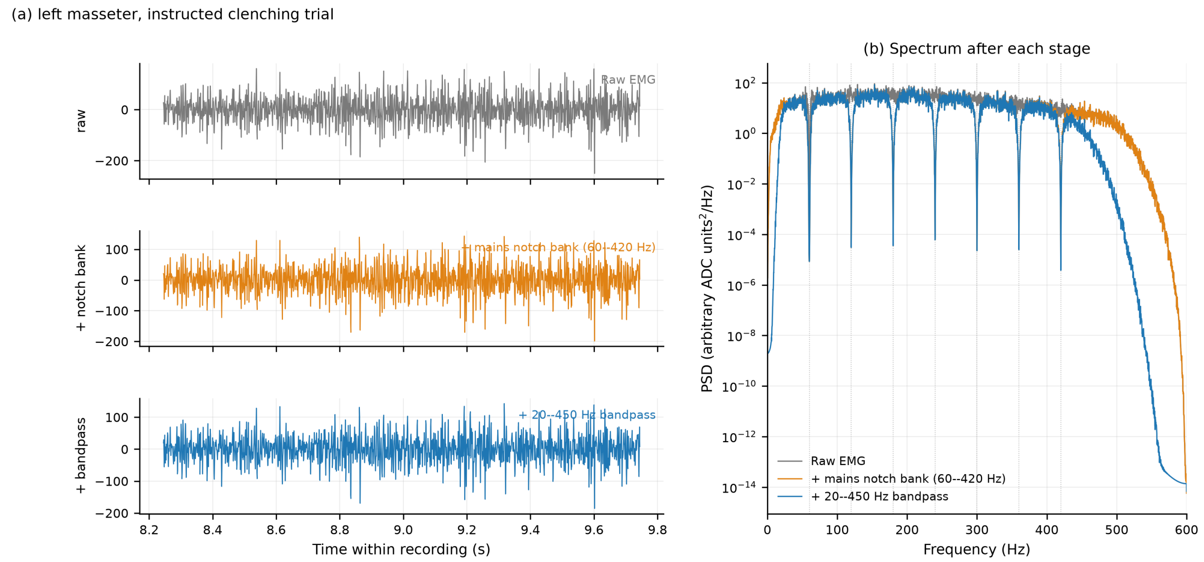

Fig. 4 Production preprocessing on a real excerpt of an instructed clenching trial (left masseter channel). (a) Time domain after each stage. (b) Power spectrum after each stage; the notch bank removes the seven mains multiples marked by dotted lines and the band-pass then removes content outside 20–450 Hz. The chain is applied to the continuous recording and windows are cut afterwards.

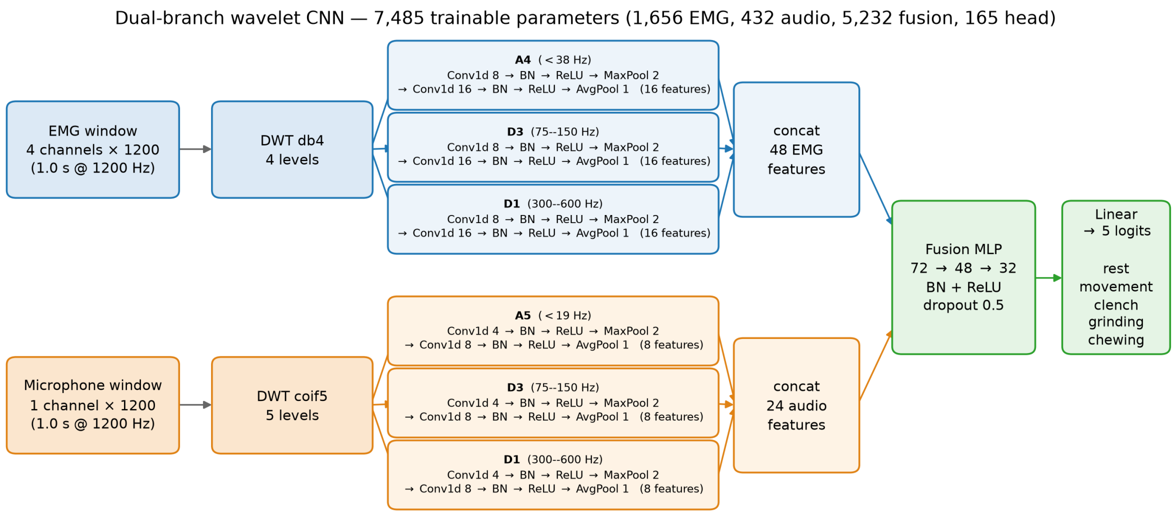

Fig. 5 Dual-branch five-class architecture. EMG and audio undergo modality-specific wavelet decomposition before processing by parallel per-band CNN sub-branches. Extracted features are concatenated and classified by a fusion MLP with outputs for rest, movement, clenching, grinding, and chewing. Layer widths, wavelet families, band edges and parameter counts are read from the instantiated model.

accuracy is dominated by chewing. The whole protocol was run with three random seeds, all of which completed all five outer folds, and metrics are computed independently per seed and then summarized. Probabilities are never averaged across seeds and no seed is selected as best.

Across the fifteen fold–seed combinations, the inner loop selected a learning rate of 1×10^{-3} every time and divided almost evenly between cross-entropy (8) and focal loss (7); the epoch budget ranged from 12 to 43 with a median of 26. Each outer fold, including its inner search, took approximately 20 minutes on the training hardware, and the complete run took 5.0 hours.

Participant-level metrics are primary. Accuracy, balanced accuracy, macro precision, recall, F1 and one-vs-rest ROC–AUC are computed within each held-out participant and then summarized across the five participants; all five individual values are reported. Pooled-window metrics are also reported, but only as descriptive summaries, because windows within a participant and recording are correlated. AUC was computed from the saved held-out class probabilities, never from hard labels or a confusion matrix. A class absent from a fold receives no AUC rather than an imputed value. No window-level bootstrap intervals or p-values are reported: the windows are clustered and overlapping, and with five participants such tests would overstate precision without supporting a population-level claim. Held-out probabilities are uncalibrated (expected calibration error 0.046); they must not be read as probabilities, but every AUC remains valid because AUC is rank-based and unaffected by any monotone rescaling.

Two secondary analyses were computed post hoc from the same held-out five-class predictions and are labeled as such. First, chewing windows were removed and the four remaining probabilities renormalized, giving a four-class problem over rest, movement, clenching, and grinding. Second, predictions and probabilities were collapsed into instructed clenching/grinding versus rest/movement/chewing. Both re-read one model rather than training a new one. A separately trained four-class model without chewing was subsequently run as part of the modality ablation below, and it agrees with the post-hoc analysis closely enough to show that the renormalization is not doing the work. The second analysis provides a binary comparison against quiet rest and other instructed activities, but it is still not a clinical bruxism-detection analysis because the positive tasks were elicited and the background class does not include unrestricted free-living behavior.

3.9 Modality ablation

RQ2 was addressed by a separate confirmatory run in which the modality was the only thing that varied. The protocol below does isolate the modality; the microphone channel it operates on does not cooperate, for reasons established after the run and reported in Section 4.2. Three conditions—audio only, EMG only, and the fused pair—were each trained on two label sets: the five-class task and the same task with chewing removed. All six conditions share the identical windows, outer folds, inner folds, three seeds, class-weighting scheme, augmentation policy and normalization scope described above, and each condition's epoch budget was selected inside its own folds by the same median rule. The unused branch is removed rather than zeroed, as described in the architecture section.

One protocol element differs from the primary run and is stated here because it explains a small level difference in the tables that follow. The ablation fixes the loss and learning rate at focal loss ($\gamma = 1.5$) and 1×10^{-3} instead of selecting them inside each fold, so that no condition can win by drawing a better hyperparameter. Under that fixed configuration the fused model reaches 80.9% accuracy and 72.8% participant-level macro-F1, against 82.1% and 73.6% for the primary run, which additionally searched the loss and learning rate. Every modality *difference* quoted below is computed within the ablation, between conditions that differ only in modality; the primary run is never differenced against an ablation condition.

4 Results

The five-class, secondary and participant-level results below come from one nested-LOSO run of the fused dual-branch wavelet CNN on the corrected signal (five outer folds × three seeds; 6,173 held-out window predictions per seed, 18,519 in total). The modality ablation comes from a second run of six matched conditions on the same windows and folds (78,399 held-out predictions in total). Every value was recomputed from the saved prediction ledgers. Results obtained before the filter correction are superseded and are not restated for comparison.

Table 2 Hyperparameters. Values for the dual-branch CNN are those of the reported run; the loss and learning rate were selected inside each fold from the stated search space. The remaining rows describe the prespecified configurations of the architecture comparison, which has not yet been run on the corrected signal. Those three models receive the identical EMG and audio pair and a fixed training configuration (AdamW, 10^{-3}, focal loss with $\gamma = 1.5$, batch size 64, balanced class weights), so the comparison isolates the architecture.

Method	Hyperparameter	Value
Wavelet dual-branch CNN	Optimizer	AdamW
	LR search space	$\{3, 10\} \times 10^{-4}$
	LR selected	10^{-3} (15/15)
	Weight decay	1×10^{-4}
	Dropout	0.5
	Batch size	64
	Loss search space	focal, cross-entropy
	Focal-loss γ	1.5
	Epoch budget	Median inner best
Dual-branch temporal CNN	Wavelet front end	As above
	Pooling per band	Mean, SD, max
	Modulation spectrum	6 bins
	Dropout	0.5
Early-fusion CNN (raw)	Conv. channels	16, 32, 64
	Kernel size	7
	Pool size	4
	Dropout	0.5
Bidirectional LSTM	Hidden dimensions	32
	Number of layers	1
	Input decimation	8×
	Dropout	0.5

4.1 Five-class performance

Averaged over the five held-out participants, the fused model achieved 82.1% accuracy, 79.1% balanced accuracy, 73.6% macro-F1, and a macro one-vs-rest ROC–AUC of 0.962 (Table 3). The three seeds agreed closely (accuracy 81.1%, 82.8% and 82.2%; macro-F1 71.7%, 74.6% and 74.5%; macro AUC 0.960, 0.963 and 0.963), so seed-to-seed variation is small relative to the variation between participants. Pooling held-out windows across participants instead—a descriptive summary only, because the windows are correlated—gives 81.0% accuracy, 70.5% macro precision, 77.9% macro recall, 72.3% macro-F1 and a macro AUC of 0.942.

For the newly added rest class, the fused model achieved 74.4% precision, 92.5% recall, an F1 of 82.2%, and a one-vs-rest AUC of 0.960 (Table 5). Rest is therefore the second-best-separated class after chewing, and the best-recalled class of the five, which is the outcome the class was added to test: activity windows can now be scored against a quiet non-event condition rather than only against one another.

The baseline comparison specified for RQ3—the dual-branch temporal CNN, an early-fusion CNN on the raw signals, and a bidirectional LSTM, all receiving the identical EMG and audio pair—has not been run since the filter correction, and the earlier numbers were measured through the defective chain, in which mains harmonics carried the majority of the in-band power. They are therefore withdrawn rather than reproduced, and Table 3 marks those rows as pending. The only comparison currently available on the corrected signal comes from a screening harness that fits one model per outer fold on 35 band-energy features, with no nested hyperparameter selection and a feature set chosen after seeing these five participants: it is directionally informative and optimistic in level. On identical windows and folds it reached 67.9% macro-F1 at 79.7% accuracy for logistic regression and 64.6% at 77.7% for gradient boosting. Seven of those 35 features are microphone band energies, and therefore read the channel described in Section 4.2; removing them and refitting on the same windows and folds gives 63.9% at 78.3% for logistic regression and 61.3% at 74.8% for gradient boosting, so the microphone features are worth 4.0 and 3.3 points of screening macro-F1 respectively and the quoted values are inflated by that much. All four numbers sit below the network's 73.6% participant-level macro-F1, but the comparison is not like-for-like—the network also receives the microphone—and it does not settle RQ3 in either direction.

Per-class one-vs-rest ROC and precision–recall curves are shown in Fig. 6. The ordering is consistent across both views: chewing and rest separate well, whereas movement combines a high AUC with a low average precision—0.914 and 0.553 respectively for the seed drawn—reflecting its small support of 333 of 6,173 windows rather than an inability to rank it.

4.2 A second measurement-chain defect, in the microphone channel

The interference audit that produced the corrected filter chain was applied to the electromyographic channels only. Repeating it on the microphone channel, after the modality ablation had

Table 3 Five-class performance, averaged over the five held-out participants and three seeds. Classes are rest, movement, clenching, grinding, and chewing. Values describe this five-participant dataset and are not population-level superiority tests. Baseline rows await the confirmatory sweep on the corrected signal; the screening rows come from a single-fit harness and are reported for direction only.

Method	Accuracy	Macro prec.	Macro rec.	Macro F1	Macro AUC
Wavelet dual-branch CNN (fusion)	**82.1%**	**70.5%**	**77.9%**	**73.6%**	**0.962**
Logistic regression, band energies[a]	79.7%	–	–	67.9%	–
Gradient boosting, band energies[a]	77.7%	–	–	64.6%	–
Dual-branch temporal CNN			*pending re-measurement*		
Early-fusion CNN, raw signals			*pending re-measurement*		
Bidirectional LSTM			*pending re-measurement*		

Macro precision and recall for the CNN are pooled-window values; accuracy, macro-F1 and macro AUC are participant-level means.
[a]Screening harness: one model fit per outer fold, no nested hyperparameter selection, feature set chosen after seeing these participants. Optimistic in level and not a confirmatory baseline.

Table 4 Modality ablation on the corrected signal, reported as run. All conditions share the same windows, folds and seeds; the unused branch is removed rather than zeroed. Accuracy, macro F1 and macro AUC are participant-level means over the five held-out participants and three seeds. Rest AUC is the one-vs-rest AUC for the quiet-rest class over pooled held-out windows, matching the convention of Table 5. Rest false alarm is the share of true rest windows classified as clenching or grinding; lower is better. **Every row that reads the microphone must be interpreted through Section 4.2**: the microphone waveforms are shared between participants, so the audio-only and fused conditions were not evaluated leave-one-subject-out on that channel, and these values bound what a duplicated, unaligned, envelope-band channel contributes rather than measuring a microphone. The EMG-only rows are unaffected.

Task	Configuration	Accuracy	Macro F1	Macro AUC	Rest AUC	Rest false alarm
Five-class	Audio only	60.8%	42.8%	0.780	0.950	16.9%
	EMG only	81.7%	72.0%	0.958	0.958	7.7%
	EMG + audio[a]	**82.1%**	**73.6%**	**0.962**	**0.970**†	**6.1%**†
Chewing removed (four-class)	Audio only	52.4%	45.0%	0.745	0.883	16.1%
	EMG only	77.7%	75.8%	0.942	0.973	7.4%
	EMG + audio[b]	**78.6%**	**76.2%**	**0.947**	**0.985**	**6.7%**

[a]Accuracy, macro F1 and macro AUC in this row are those of the primary nested-LOSO run of Table 3, which additionally selected the loss and learning rate inside each fold, and are quoted here so that one fused number appears throughout the paper. Cells marked † have no counterpart in that run and are taken from the ablation's own fused condition, which under the fixed training configuration scored 80.9% accuracy, 72.8% macro F1 and 0.963 macro AUC. **Every modality difference quoted in the text is computed inside the ablation**, between conditions that differ only in modality, and never by subtracting an ablation row from this one; those matched differences are +0.009 macro-F1 on the five-class task and +0.004 with chewing removed, both smaller than the spread between seeds.
[b]Separately trained on the four remaining classes, not the post-hoc renormalization reported in Table 6.

already been run, revealed a defect of a different kind, and it is reported here for the same reason the first one was.

Fingerprinting each channel by a rotation-invariant signature of its samples shows that the 100 recordings contain 100 distinct waveforms on each of the four EMG channels and only 37 on the microphone channel. Eighty-three recordings carry a microphone waveform that is bit-identical, after a circular rotation of between 0.2 and 8 s, to that of another participant's recording of the same condition, and all four of the S01–S04 quiet-rest recordings share a single waveform; the identity is confirmed by exact array comparison after the recovered rotation, not by correlation. The channel is also unaligned with the electromyogram—over the 45 chewing recordings, where every muscle burst coincides with an impact, the envelope cross-correlation at zero lag has a median of −0.017 and peaks at a median

Table 5 Per-class performance of the five-class fused model on held-out windows, averaged over three seeds. AUC is one-vs-rest, computed from held-out probabilities.

Class	Precision	Recall	F1	AUC	Windows
Rest	74.4%	92.5%	82.2%	0.960	590
Movement	34.9%	75.9%	47.8%	0.934	333
Clenching	74.3%	64.0%	68.7%	0.921	799
Grinding	72.1%	71.6%	71.8%	0.920	816
Chewing	97.0%	85.4%	90.8%	0.974	3,635
Macro average	70.5%	77.9%	72.3%	0.942	6,173

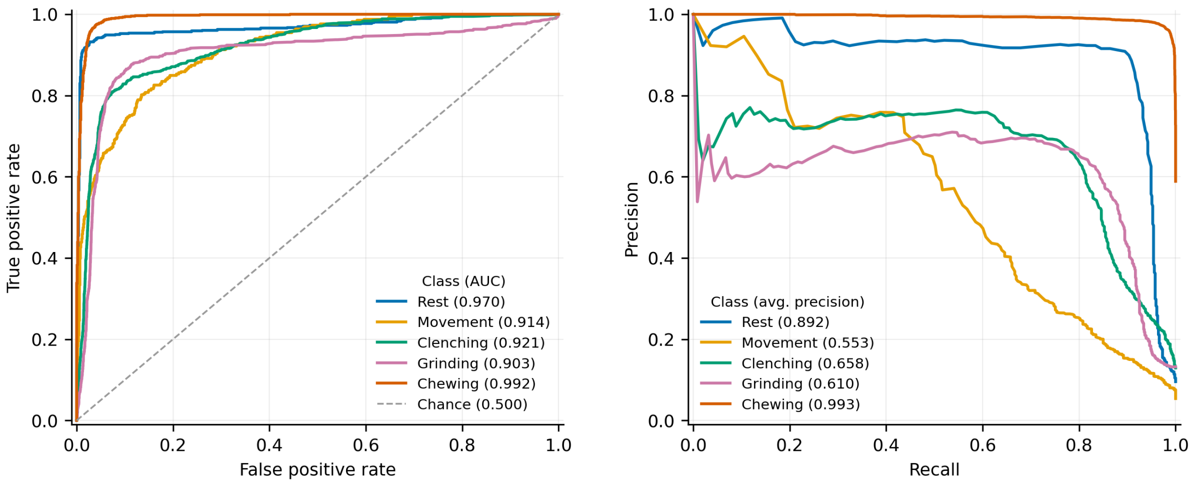

Fig. 6 One-vs-rest ROC (left) and precision–recall (right) curves for the five experimental classes, pooled over the held-out windows of seed 0 (the lowest seed index, a fixed rule rather than a choice made after seeing the scores). Each curve depicts one trained model per fold, so the legend values are that seed's; the three-seed means are in Table 5. Precision–recall separates the classes far more sharply than ROC does, because it is sensitive to the very unequal class sizes.

absolute lag of 18 s—and 96% of its power lies below 10 Hz (range 91.4–98.7%), so it behaves as a rectified sound-level output rather than as an acoustic waveform. It is integer-valued with a 1.0-count quantization step, and the 20-Hz high-pass of the production chain retains a median of 1.19% of its variance; for the clenching and grinding conditions what remains is at or below the quantization floor. The three `.npy` acquisition companions that exist reproduce the electromyographic and trigger columns exactly and contain an all-zero microphone column, and the video files carry no audio stream, so no second copy of the signal exists to recover.

The consequence for Table 4 is specific, and it is not a question of statistical power. Every number in that table is a correct report of what the run produced under the protocol described above, and the electromyographic conditions are unaffected: the EMG channels are distinct across every recording, and the five-class, secondary and participant-level results elsewhere in this paper do not read the microphone at all. But because a held-out participant's microphone waveform is, for four of the five participants, already present in that fold's training data, the audio-only and fused conditions were not evaluated leave-one-subject-out on the audio channel. The numbers therefore bound what a duplicated, unaligned, envelope-band channel can contribute under this architecture; they do not measure what a microphone contributes, and the direction of the bias is not uniform—where the duplicated waveform carries the matching condition label it is optimistic, and where it does not, as in the one recording whose microphone waveform belongs to a different condition entirely, it is pessimistic. We therefore report the ablation as it was run, withdraw the reading that the microphone is a complementary channel whose value lies in separating quiet rest from tooth contact, and record RQ2 as unanswered. Notably, the one participant whose microphone waveforms are largely unique—P5, with 4 of 20 recordings duplicated against 19–20 of 20 for the others—is the weakest of the five on every audio-only metric and the strongest on every EMG-only metric, which is the pattern this defect predicts and which is not otherwise explained.

Two properties of this defect generalize beyond this dataset, and they are the reason it is reported at length rather than noted in the limitations. First, it is invisible from any single recording: every affected file is individually well-formed, correctly sized, and carries plausible values. It becomes obvious only when two files are compared, and a data audit that examines recordings one at a time—as ours did, and as most do—cannot see it. Second, the audit that caught the mains defect existed, was thorough, and was pointed at one modality. We now compute a rotation-invariant fingerprint of every channel of every recording, group them, and refuse a run whose held-out participant's signal appears in its own training set; the check costs one pass over the data and would have caught this a year earlier.

4.3 Modality ablation (RQ2)

The confirmatory modality ablation has now been run on the corrected signal, and its results are summarized in Table 4 and Fig. 7. Every value below must be read against Section 4.2: the microphone channel of this collection is duplicated across participants, so these conditions bound what such a channel contributes rather than measuring the microphone.

Audio alone is clearly informative, though not by the measure a reader would reach for first. Its accuracy of 60.8% is essentially the 60.5% a classifier that always predicted chewing would score on the same participants, so accuracy carries no information here at all; the statistics that do are the participant-level macro-F1 of 42.8%, against 16.5% for a uniform guess, and the macro one-vs-rest AUC of 0.780, against 0.5 for chance—both obtained from a single microphone and 3,525 parameters. That information is not spread evenly over the classes (Fig. 7c). From sound alone the model ranked quiet rest at an AUC of 0.950, within 0.008 of what the EMG branch reaches for that class, and chewing at 0.899, while movement sat at 0.427—at or below chance. We previously read this as the microphone hearing whether the mouth is quiet and whether the participant is eating. Section 4.2 withdraws that reading: all four S01–S04 quiet-rest recordings share one microphone waveform, so a fold that holds out one of them has already been trained on that exact rest audio three times, and a single scalar—the broadband amplitude of the filtered window, with no learning at all—recovers most of this discrimination (rest 0.852, chewing 0.865, clenching 0.777). What the audio branch separates is recording level, not sound.

Audio alone is nonetheless far weaker than EMG alone, which reached 81.7% accuracy and 72.0% macro-F1 on the same windows, and it is not a substitute: 16.9% of quiet-rest windows were read as clenching or grinding by the audio-only model, against 7.7% for EMG only.

Adding audio to EMG produced no overall improvement that this dataset can resolve. Averaged over the five participants and three seeds, the fused condition scored 72.8% macro-F1 against 72.0% for EMG only; with chewing removed the two were 76.2% and 75.8%. Computed before rounding, the paired mean differences are +0.009 and +0.004 macro-F1, and both are smaller than the variation between training seeds. The per-seed five-class differences were −0.055, +0.042 and +0.039 macro-F1, and the paired per-participant differences ranged from −0.106 to +0.104 (Fig. 7b). Three of five participants improved on each task, and the participant ordering was not preserved between the two tasks—the participant who lost the most on the five-class task gained the most once chewing was removed—so the between-participant spread is best read as noise rather than as a stable per-person effect. On this evidence we do not claim a macro-F1 benefit from fusion.

Two effects were nevertheless consistent across seeds, and both concern the quiet-rest class. The share of quiet-rest windows classified as clenching or grinding fell from 7.7% under EMG only to 6.1% under fusion, a 21% relative reduction, in every seed on both label sets and with non-overlapping ranges on the five-class task (7.1–8.5% against 5.3–6.6%; Fig. 7d); and the one-vs-rest AUC for quiet rest rose from 0.958 to 0.970 on the five-class task and from 0.973 to 0.985 with chewing removed.

We reported these as the microphone's specific contribution, and we withdraw that. Consistency across seeds is the wrong evidence here, because it is exactly what a duplicated channel produces: the four S01–S04 rest recordings share one waveform, so every seed of every fold is fitting and scoring the same rest audio, and the effect is reproducible for a reason that has nothing to do with a microphone. The specific hypothesis embedded in RQ2—that the audio benefit is concentrated in chewing—cannot be evaluated at all on this channel, and neither can its negation. What the ablation now supports is bounded: under this architecture, a channel of this kind neither carries the task alone nor raises the headline macro-F1, at a cost of 1,584 parameters. Whether a working microphone would is untested. Section 4.2 states what a future collection must do differently for the question to be answerable.

4.4 Secondary analyses with rest

When chewing was excluded, the remaining four-class problem comprised rest, movement, clenching, and grinding. Averaged over the held-out participants, this analysis yielded 79.8% accuracy, 77.9% macro-F1, and a macro AUC of 0.944; the corresponding pooled-window values were 77.1% accuracy, 76.6% macro precision, 78.4% macro recall, 77.2% macro-F1, and a macro AUC of 0.917 (Table 6). Removing chewing lowers accuracy by 2.3 percentage points but *raises* macro-F1 by 4.3 points, because the metric is no longer dominated by a single large, easily separated class. Reporting both prevents the comparatively recognizable chewing class from carrying the global summary. A four-class model trained separately on the same four classes, run as part of the modality ablation, reaches 78.6% accuracy, 76.2% macro-F1 and a macro AUC of 0.947. That it lands so close to the post-hoc values confirms that those values are not an artifact of the renormalization shortcut.

For a task-level binary analysis, clenching and grinding were treated as the positive group and rest, movement, and chewing as the comparison group. This yielded 93.6% accuracy, 90.6% precision, 84.3% sensitivity, 96.9% specificity, an F1 of 87.3%, and a ROC–AUC of 0.966 on pooled windows, and 94.0% accuracy with a participant-level macro-F1 of 90.9% and AUC of 0.977. Within the quiet-rest condition, 6.0% of rest windows (35 of 590 for one seed and 36 of 590 for each of the other two) were classified as clenching or grinding. This analysis includes a laboratory non-event class but still does not measure spontaneous bruxism detection in a continuous free-living stream: a 6% per-window false-alarm rate at a 0.5-s decision interval would correspond to roughly 440 false alarms per hour if the background were quiet rest throughout, which is precisely the quantity a deployable detector must report and this design cannot estimate.

4.5 Class-level diagnostics

The highest per-class F1 was observed for chewing (90.8%) and the lowest for movement (47.8%). Movement's low F1 is a precision problem rather than a detection problem: its recall is 75.9% and its one-vs-rest AUC is 0.934, but its precision is 34.9% because it is the smallest class (333 windows, 5.4% of the total) and absorbs 10.3% of the 3,635 chewing windows, which are eleven times more numerous. Clenching and grinding were most often confused with each other: 19.3% of clenching windows were labeled grinding and 13.5% of grinding windows were labeled clenching, which together account for the majority of both classes' errors. Rest had the highest recall of any class at 92.5%, and its dominant error was into grinding (5.9%). Confusion between quiet rest and the combined instructed clenching/grinding group therefore occurred in 6.0% of rest windows. These errors matter most, because they approximate false alarms under the limited quiet-rest condition.

We used t-distributed stochastic neighbor embedding (t-SNE) [22] to visualize the 72-dimensional fusion features of held-out windows (Fig. 8). Chewing occupies several large, well-separated regions, consistent with its high AUC and with food-dependent acoustic and rhythmic structure. Rest forms a small number of tight, compact islands, which is what a low-variance quiet condition should look like. Clenching and instructed grinding interleave repeatedly, forming mixed regions rather than separate territories,

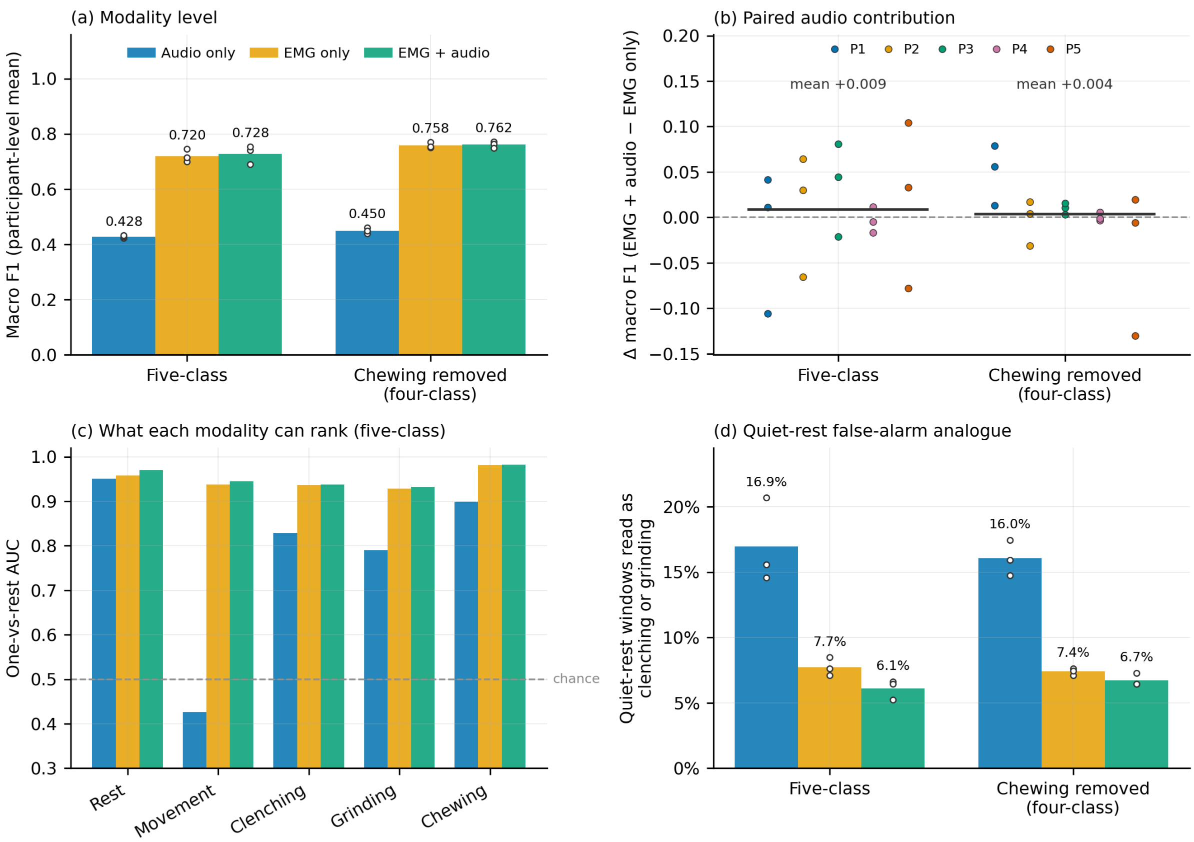

Fig. 7 Modality ablation on the corrected signal. Every condition shares the same windows, outer and inner folds, seeds and fixed training configuration; only the modality differs, and the unused branch is not constructed. (a) Participant-level macro-F1 for the five-class task and for the same task with chewing removed; bars are the mean over five participants and three seeds, open markers the three seed means. (b) The paired fusion-minus-EMG-only difference for every participant and seed, which is the spread the bars in (a) hide: the mean is close to zero on both tasks and the individual differences straddle it in both directions. (c) One-vs-rest AUC per class on the five-class task. Audio alone ranks quiet rest almost as well as EMG does and chewing appreciably less well, but sits at chance for movement. (d) Share of quiet-rest windows classified as clenching or grinding—the closest available analogue of a false alarm—which falls in every seed and on both label sets when the microphone is added. **Panels (a)–(d) are reported as run and must be read through Section 4.2.** The microphone waveforms are shared between participants—all four S01–S04 quiet-rest recordings are one waveform—so the audio-only and fused conditions were not held out on that channel, and the seed-to-seed consistency in (d) is what a memorized constant produces rather than evidence of a modality effect. The EMG-only condition is unaffected.

and movement windows appear mostly as a scatter along the borders of the larger classes rather than as a territory of their own. The embedding therefore reproduces the two error patterns visible in the confusion matrix. Because t-SNE is an exploratory nonlinear visualization whose apparent distances are not metric, these clusters are not evidence of clinical phenotypes and no quantitative claim rests on them.

The held-out confusion matrix is shown in Fig. 9. For the seed shown, the diagonal counts were 549 of 590 rest windows, 235 of 333 movement, 531 of 799 clenching, 553 of 816 grinding, and 3,110 of 3,635 chewing. The corresponding class supports are listed in Table 5.

4.6 Participant-level results

Table 7 reports five-class results for each held-out participant. Across participants, accuracy ranged from 72.0% to 91.7%, macro-F1 from 64.5% to 82.6%, and macro AUC from 0.896 to 0.985 (Fig. 10). The spread is substantial relative to the mean and is not explained by class balance alone.

Table 6 Secondary analyses computed post hoc from the held-out five-class predictions, averaged over three seeds. These re-read one trained model; they are not separately trained tasks.

Analysis	Accuracy	Precision	Recall	Specificity	F1	AUC
Excluding chewing[a]	77.1%	76.6%	78.4%	–	77.2%	0.917
Collapsed binary[b]	93.6%	90.6%	84.3%	96.9%	87.3%	0.966

Values are pooled over held-out windows. The corresponding participant-level means are 79.8% accuracy / 77.9% macro-F1 / 0.944 AUC when chewing is excluded, and 94.0% accuracy / 90.9% macro-F1 / 0.977 AUC for the collapsed binary analysis.
[a]Macro precision, recall, F1, and one-vs-rest AUC across rest, movement, clenching, and grinding, after removing chewing windows and renormalizing the remaining four probabilities.
[b]Instructed clenching/grinding treated as positive; rest/movement/chewing treated as the comparison group. This is task discrimination, not clinical diagnosis.

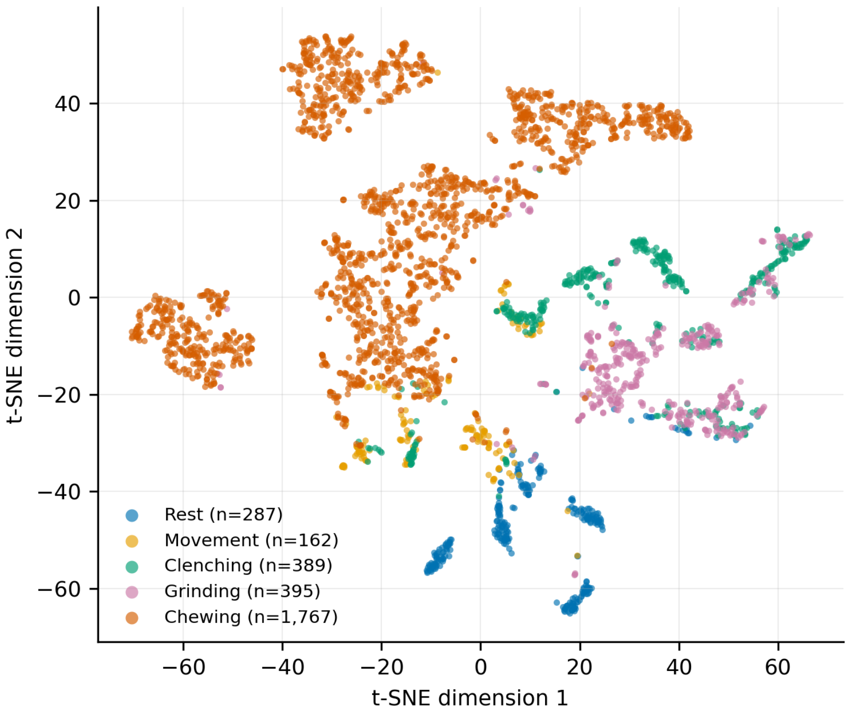

Fig. 8 t-SNE visualization of the 72-dimensional learned fusion features for held-out windows, colored by experimental class. Chewing occupies large separate regions and rest forms tight compact islands, whereas clenching and instructed grinding interleave and movement scatters along class borders. Embeddings were recomputed from the saved fold checkpoints using only outer-test windows. This exploratory visualization does not establish clinical phenotypes.

Two participants illustrate why accuracy and AUC should be read together. Participant 4 had the lowest accuracy (72.0%) but the second-highest macro AUC (0.982): its held-out windows are ranked well by the model's scores, and the loss is concentrated in where the decision boundaries fall for that participant rather than in the model's ability to order its windows. Participant 1 shows the opposite pattern—the lowest AUC (0.896) and the lowest macro-F1 (64.5%), at an accuracy no better than participant 4's—and it also contributes the largest single share of windows (1,692 of 6,173), so it dominates any pooled-window summary while counting as one of five in the participant-level summary. This is one concrete reason the participant-level aggregation is treated as primary here.

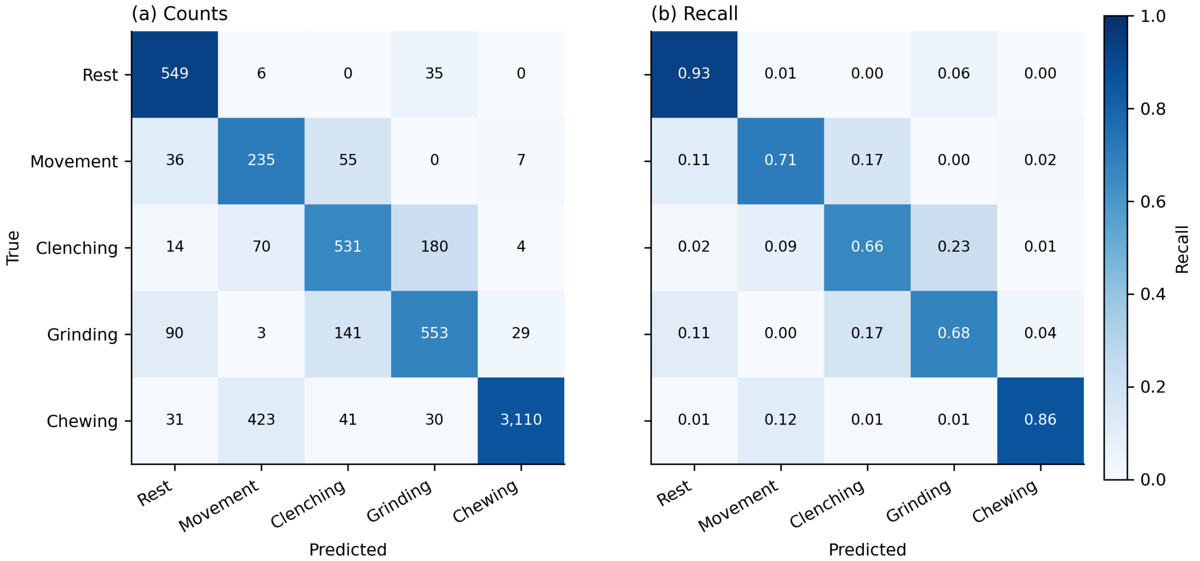

Fig. 9 Held-out confusion matrix for rest, movement, clenching, grinding, and chewing, pooled across the five leave-one-subject-out folds of seed 0 (selected by the fixed lowest-seed-index rule, not by score). (a) Counts. (b) Row-normalized recall. The two dominant error patterns are the mutual clenching–grinding confusion and the leakage of chewing windows into the much smaller movement class.

These five values describe internal LOSO behavior only. They do not estimate performance across age groups, anatomical variation, dentition, bruxism phenotypes, or clinical settings, and with five participants the standard deviations below describe the sample rather than a sampling distribution.

Table 7 Five-class performance for each held-out participant, averaged over three seeds.

Held out	n	Acc.	Macro F1	Macro AUC
Participant 1	1,692	72.7%	64.5%	0.896
Participant 2	1,007	91.7%	74.4%	0.977
Participant 3	1,133	86.1%	74.7%	0.971
Participant 4	1,167	72.0%	71.8%	0.982
Participant 5	1,174	87.8%	82.6%	0.985
Mean	6,173	82.1%	73.6%	0.962
SD	—	9.1	6.5	0.037

4.7 Training behavior and computational measurements

Because the final model of each fold is refit for a fixed epoch budget chosen inside that fold, there is no early stopping and no validation set at refit time; the curves in Fig. 11 are training-fit diagnostics, and the held-out participant appears in none of them. Budgets ranged from 12 to 43 epochs (median 26) across the fifteen fold–seed combinations, and training loss had largely flattened by the end of each budget. The gap between the final training-fit macro-F1 and the corresponding held-out macro-F1 averaged 0.133 and ranged from 0.022 (participant 5) to 0.269 (participant 1)—the same participant whose windows are the most numerous and the least well ranked. That ordering is the expected signature of between-participant variation rather than of optimization failure.

The five-class model contained 7,485 trainable parameters and occupies 29.2 KiB at 32-bit precision. Three latencies must be kept separate. The *input-context latency* is 1000 ms: the window length, and therefore the earliest point at which any decision about an event can exist. The *decision update interval* is 500 ms, the stride. The *processing latency*—filtering, wavelet transform and forward pass for one pre-segmented window on an Intel CPU—is 2.97 ms (2.88 ms forward pass, 95th percentile 2.95 ms, plus 0.08 ms preprocessing), giving an earliest possible decision at 1003 ms. For reference on the same bench, an early-fusion CNN baseline (19,141 parameters) took 0.29 ms and a bidirectional LSTM (10,309 parameters) 1.39 ms.

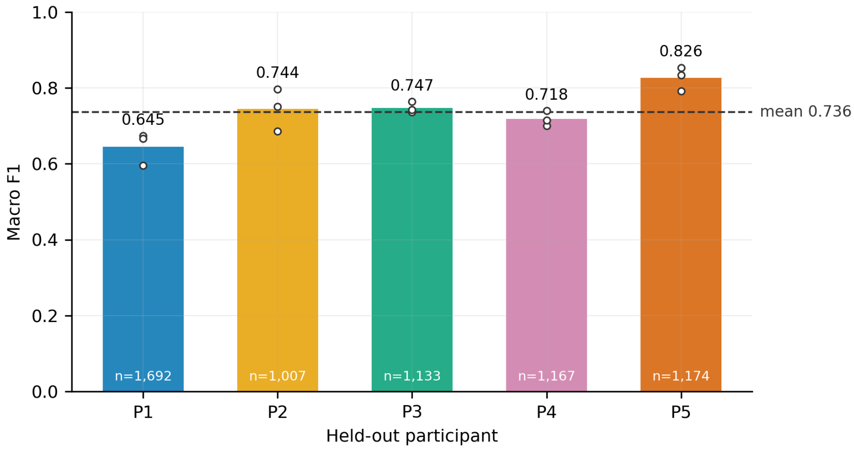

Fig. 10 Macro-F1 for each held-out participant. Each bar is the mean over the three seeds of that participant's own leave-one-subject-out fold, the open markers are the three individual seeds, and the label inside each bar gives that participant's held-out window count. Five participants cannot characterize population variability, so no generalization claim follows from the mean.

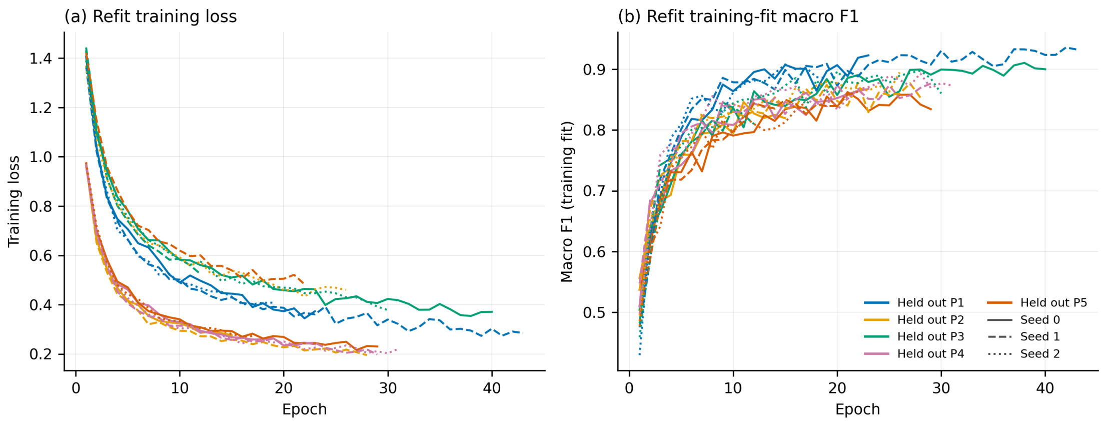

Fig. 11 Refit training loss (a) and training-fit macro-F1 (b) for all fifteen fold–seed combinations, colored by held-out participant. Line style encodes the seed: solid seed 0, dashed seed 1, dotted seed 2. Each refit runs for the epoch budget chosen inside its own fold, so the curves end at different epochs. The held-out participant contributes to no curve and is scored once, after training ends.

The compute time alone is not detection latency, and none of these measurements supports a wearable or real-time claim. No streaming implementation exists, the production filter chain is zero-phase and therefore acausal, and the measurements were taken on laptop and desktop hardware. Power consumption, end-to-end latency, wireless synchronization, and reliability on a wearable device remain unmeasured.

5 Discussion

5.1 Engineering interpretation

Within this dataset, the dual-branch wavelet CNN separated five instructed experimental conditions with a participant-level macro-F1 of 73.6% and a macro one-vs-rest AUC of 0.962, and it did so under a protocol in which no held-out participant's windows touched normalization, class weighting, augmentation, hyperparameter selection, or training. We do not claim that it outperformed the alternative architectures: that comparison has not been run on the corrected signal, and the only interim evidence—a single-fit screening harness on band energies—puts logistic regression at 67.9% macro-F1 against the network's 73.6% on identical windows and folds, a gap small enough at $N = 5$ that it cannot carry an architectural claim in either direction.

Adding quiet rest changed the task from classification among activity windows to discrimination of five experimental conditions including a limited non-event condition, and rest proved to be the most reliably recovered class (92.5% recall, AUC 0.960). The residual 6.0% of rest windows assigned to clenching or grinding is the most decision-relevant number in the paper, because it is the closest available analogue of a false alarm.

The audio contribution (RQ2) is not measured by this study, and the reason is a property of the channel rather than of the sample size. The previously reported figure of +0.035 macro-F1 was obtained through the defective filter chain and is withdrawn; the matched ablation on the corrected signal puts the paired mean contribution at +0.009 macro-F1 on the five-class task and +0.004 once chewing is removed, neither exceeding the seed-to-seed spread. Both of those numbers, however, were computed on a channel that carries other participants' audio in four folds out of five (Section 4.2), so they bound rather than estimate, and they bound a duplicated envelope-band channel rather than a microphone. The one statement that survives intact is about EMG: it accounts for essentially all of the classification performance reported in this paper, and that conclusion does not depend on the microphone at all.

We previously offered three reasons to keep the microphone, and two of them do not survive. That sound alone separated quiet rest at an AUC of 0.950 and chewing at 0.899 is not evidence about sound: those recordings share waveforms across participants, and a single scalar amplitude feature recovers most of the same discrimination. That the quiet-rest false-alarm share fell from 7.7% to 6.1% in every seed is not evidence either, for the same reason and more sharply—all four S01–S04 rest recordings are one waveform, so seed-to-seed consistency is the signature of a memorized constant rather than of a real effect. The third reason stands, because it is a statement about the model and not about the data: the pathway costs 1,584 parameters and 6.2 KiB, so carrying a microphone would not trade against model size if one were carried. Whether it should be is a question this study leaves open rather than answers weakly, and the difference matters, because a weak positive would justify the sensor and an untested one does not.

Figure 12 shows the traces that first suggested a role for audio. Quiet rest is quiet in both channels; instructed grinding produces sustained electromyographic activation with comparatively little microphone energy; chewing produces rhythmic bursts alongside large microphone excursions, whose median filtered RMS across the dataset is 3.3 times that of instructed grinding. That contrast is real and it is measurable, but it is a contrast in a 1–3-Hz envelope rather than in an acoustic transient, and the microphone trace shown is not synchronous with the electromyographic trace beside it. We read those traces as showing that audio would mainly help identify eating; the figure could not have shown that, and neither the ablation nor any other analysis in this paper tests it.

We therefore make no claim about fusion in either direction. The claim we do make is about method: a modality contrast is only as good as the modality, and the check that establishes whether a channel is what it is labelled as costs one pass over the data.

5.2 What the filter correction changed, and why it is reported

The single largest determinant of the results in this paper was not the architecture, the loss function, or the fusion strategy. It was whether the mains harmonics were removed. Run over the identical 6,173 windows and folds, the screening harness scored 39.3% macro-F1 (56.0% accuracy) through the superseded chain and 67.9% (79.7%) through the corrected one—a step larger than every other change we evaluated combined—and participants whose held-out accuracy had previously sat at or below the five-class chance level moved into the normal range.

The mechanism is easy to state and easy to miss. A chain that notches only the supply fundamental is correct when the fundamental is the dominant interference. Here it was not: the acquisition hardware had already removed it, and what remained was harmonic. A filter-response diagram of that chain looks textbook-correct, and the pipeline was reproducible, version-controlled, and tested; none of that could detect the error, because nothing in the figure set placed the filter response over the measured spectrum of the recordings. When a classifier is trained on such a signal, between-participant differences in interference amplitude can dominate between-condition differences in physiology. In our recordings, correction reduced the between-participant amplitude spread from 5.0× to 3.0× and eliminated all 20 participant pairs in which one person's resting EMG had exceeded another person's active EMG. A model trained before the correction could reduce its loss by learning who a window came from rather than what they were doing, and leave-one-subject-out evaluation is exactly the protocol that then reports the consequence as poor generalization.

We report this at length for three reasons. First, it is the honest provenance of the numbers: the results in this paper are not an incremental improvement on our earlier ones; they replace them, and the two sets are not comparable. Second, the vulnerability is not specific to this dataset. Head-mounted electrodes, long leads, and acquisition software that applies undocumented fixed filtering are common in ambulatory jaw-activity work, and published accuracies are rarely accompanied by a measured interference statistic. Third, the check is cheap: the fraction of in-band power lying within a few hertz of each mains multiple can be computed per recording in seconds, and it separated the contaminated from the clean recordings here by more than an order of magnitude. We suggest that instrumental studies of masticatory-muscle activity report such a statistic alongside their accuracies, and that filter-response figures be drawn over the measured spectrum of the data rather than alone.

The current result should be compared cautiously with the greater-than-90% accuracy reported by Sonmezocak and Kurt using EMG alone [8]. Their classes included jaw opening, clenching, rhythmic grinding, and fatigue/pain; the present classes include rest, movement, clenching, grinding, and chewing. Dataset composition, preprocessing, sample size, and validation design also differ, and our evaluation is leave-one-subject-out, so every reported number is measured on a participant the model has never seen. A within-participant or window-shuffled split of the same data would score considerably higher without measuring anything about a new user. These studies therefore answer different classification questions, and the numerical accuracies do not show that either method is clinically superior.

The clenching–grinding distinction was retained to test whether the architecture could separate static and lateral tooth-contact tasks. It is the weakest distinction in the model: one fifth of clenching windows were labeled grinding and one seventh of grinding windows were labeled clenching, and the two classes interleave in the learned feature space rather than occupying separate regions. We do not claim that this distinction is always necessary for clinical management. Both are only parts of the wider bruxism construct, and a 1-second window may be genuinely unlabelable when a grinding bout passes through a static clenching phase. The observed confusion therefore reflects both signal overlap and task-label ambiguity, and the collapsed tooth-contact analysis—in which the distinction is not required—is correspondingly stronger (93.6% accuracy, AUC 0.966).

5.3 Effect of the chewing class

Chewing was selected as a relevant functional confound, not as evidence of bruxism. In the five-class model it comprised 58.9% of the windows and achieved an F1 score of 90.8%, the highest of any class. Excluding it changed participant-level accuracy from 82.1% to 79.8% and macro-F1 from 73.6% to 77.9%. The direction of those two changes is the point: a headline accuracy computed over this class distribution is carried substantially by the easiest and largest class, whereas macro-F1 improves once that class is removed because the remaining four are more evenly sized. Reporting both analyses, and reporting macro-F1 alongside accuracy throughout, is what keeps the summary honest.

The post-hoc two-group analysis places quiet rest in the comparison group, which makes it more informative than an activity-only contrast. It must still not be labeled clinical "bruxism detection." Chewing can include tooth contact, clenching

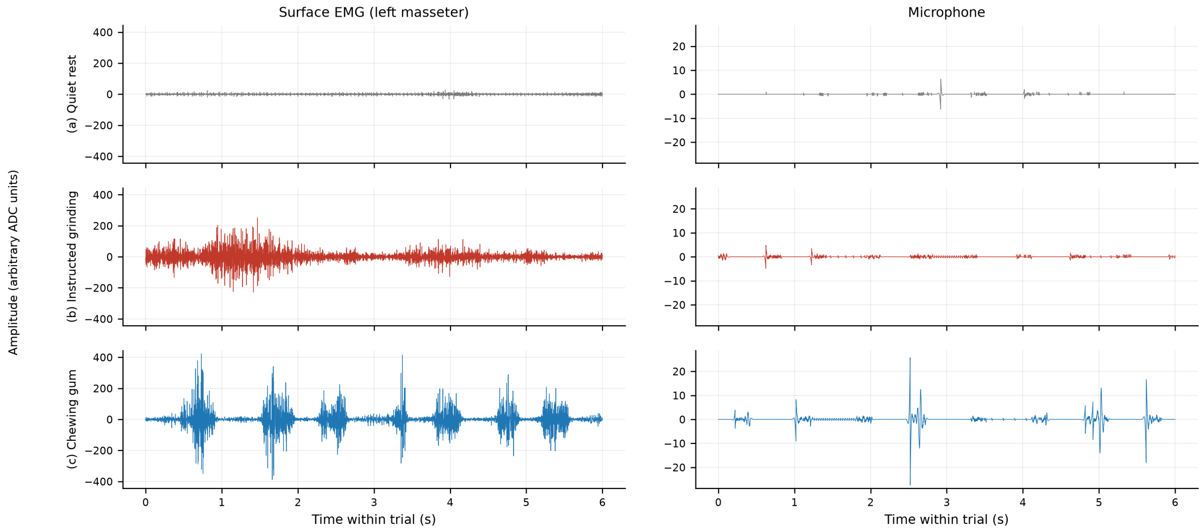

Fig. 12 Six-second excerpts of one participant's quiet rest, instructed grinding, and gum chewing, after the corrected production filter chain, on a common scale within each column. (a) Rest is quiet in both channels. (b) Instructed grinding produces sustained EMG activation without a clear burst–relaxation rhythm, and comparatively little microphone energy. (c) Chewing produces rhythmic EMG bursts separated by relaxation intervals, alongside large microphone excursions; over the whole dataset the median filtered microphone RMS during chewing is 3.3 times that during instructed grinding (4.2 times in raw counts before filtering). **The microphone row must not be read as time-aligned with the EMG row above it**, and the excursions are envelope-band rather than acoustic: see Section 4.2. Representative examples are illustrative; they are not a measurement of the modality contribution and not validation against natural bruxism episodes.

may occur without clinically important bruxism, and natural bruxism also includes bracing or thrusting without tooth contact. Quiet laboratory rest is much narrower than unrestricted everyday background behavior. A deployable detector will require continuous-stream onset and offset evaluation, diverse non-event behaviors, false alarms per unit time, and event-level sensitivity and positive predictive value.

5.4 Clinical interpretation within current bruxism and TMD frameworks

The present sensors measure muscle and sound patterns, not the biological, psychological, or social meaning of those patterns. Under the biopsychosocial model, the same motor activity may have different context, drivers, and consequences across people and time [25]. A future device should therefore be positioned as one source of instrumentally based assessment within STAB, combined when appropriate with subject report, clinical examination, psychosocial information, sleep-related factors, medications and substances, and possible consequences [2].

The manuscript also deliberately avoids claiming that bruxism caused TMD or that classifying a task can diagnose or guide treatment of TMD. INfORM/IADR emphasizes validated history and examination, patient-centered decision-making, and conservative management, and cautions that unsupported technological measurements should not replace standard assessment [27]. Any future clinical study of this system should predefine its intended use—for example, behavior quantification, biofeedback support, or screening for further assessment—and validate that use against an appropriate reference standard.

The need for precise claims is not semantic alone. Practice surveys show inconsistent assessment and management approaches [28], and studies of students and educational programs identify persistent gaps in bruxism and TMD knowledge [29, 30]. Labeling an instructed-task classifier as a diagnostic bruxism device could reinforce precisely the conflation that current terminology and education seek to correct.

5.5 Natural fluctuation and future home measurement

Laboratory repetitions do not capture the frequency, duration, context, or natural variability of awake behavior. EMA studies demonstrate the value of repeated real-time reports in the natural environment [32]; longer observations also show that behavior estimates must be interpreted across days and months rather than from a single session [33]. Bracci et al. [31] recommend selecting subject-based, clinically based, or instrumentally based metrics according to the research question. Colonna et al. [3] further illustrate the feasibility and challenges of 24-hour home EMG.

A next-stage awake study should combine multi-day audio–EMG with synchronized EMA, a clearly defined background/rest class, and blinded annotation of candidate episodes. Day-level and participant-level performance should be reported, including false positives per hour. Privacy-preserving audio features or on-device feature extraction should be considered before home deployment.

No conclusion about sleep bruxism can be drawn from the present awake laboratory tasks. A sleep study would require overnight data, appropriate sleep staging and event criteria, comparison with a suitable reference such as polysomnography or validated portable assessment, and repeated nights to evaluate night-to-night variability [34]. Separate models may be required for awake and sleep contexts.

5.6 Validity and limitations

The principal limitation is the cohort size ($N = 5$). The 6,173 overlapping five-class windows provide many optimization examples but do not replace independent participants: adjacent windows share half their samples, and a single participant contributed 27% of them. Window-level confidence intervals or tests would overstate precision and are deliberately not reported, and LOSO does not establish external generalization when only five people are available. The between-participant spread observed here—accuracy from 72.0% to 91.7% and macro-F1 from 64.5% to 82.6%—is itself the clearest evidence that a five-person mean is not a population estimate. Age, facial anatomy, subcutaneous tissue, dentition, electrode placement, skin impedance, behavior intensity, and prior diagnosis may all affect EMG and audio signals.

One result in this manuscript is incomplete rather than negative, and is marked as such. The architecture comparison (RQ3) has not been run on the corrected signal; Table 3 carries explicit pending markers, and the screening values quoted for direction come from a harness that fits one model per fold with a feature set chosen after seeing these participants, which makes it optimistic in level. The primary five-class run is complete: all three prespecified seeds finished all five outer folds, and seed-to-seed variation was small (macro-F1 71.7%, 74.6% and 74.5%) relative to the between-participant spread. The modality ablation (RQ2) ran to completion, and its mean effect is smaller than its seed-to-seed spread—but that is no longer the binding limitation. Its input channel is defective in a way no amount of running would fix and no reanalysis of these files can repair: the microphone waveforms are shared between participants, so the ablation was not a leave-one-subject-out evaluation on that channel (Section 4.2). We report it as run because the numbers are a correct record of the experiment, and we do not treat it as an answer to RQ2. The distinction between an underpowered result and an invalid input matters for what happens next: more participants would fix the first, and only a new acquisition fixes the second.

Several analysis choices remain declared but unexercised. Held-out probabilities are uncalibrated (expected calibration error 0.046); temperature scaling is implemented but was not applied, so the scores must not be read as probabilities, though every reported AUC is unaffected. A per-participant normalization scope, which uses the held-out participant's unlabeled signal distribution but never their labels, is implemented and prespecified; a screening estimate values it at roughly +0.06 macro-F1, but it is a transductive test-time adaptation that would require a per-user fitting session, and no number in this paper uses it. Similarly, averaging predictions over longer contexts raises screening macro-F1 from 0.735 at one window to 0.821 at about 5.5 seconds; because every recording here contains a single condition, that aggregation approaches a majority vote over a homogeneous trial and is valid only as a labeled trial-level secondary analysis, never as continuous-stream detection. The 1-second window is therefore a constraint of this design rather than a tuned parameter.

Construct validity is limited because the task set covers tooth-contact clenching and grinding but not bracing or thrusting without tooth contact. The participants' prior diagnoses were not independently phenotyped for this study. Criterion validity is not established because no spontaneous episode was verified against a clinical or instrumental reference standard. Ecological validity is limited because participants intentionally performed the tasks in a controlled laboratory. Elicited activities may be more regular, longer, or more forceful than natural activity, making them easier to classify. Device validity is consequently limited to internal discrimination of these instructed task windows.

Including quiet rest removes one limitation of an activity-only design, but the background data remain incomplete. Rest was recorded in a controlled laboratory and did not systematically sample speaking, swallowing, facial expression, head movement, coughing, eating outside the scripted task, environmental noise, or other free-living behaviors. The five-class analysis can estimate confusion against quiet rest—6.0% of rest windows fell into the clenching/grinding group—but it cannot estimate false alarms per hour during unrestricted use, because the denominator that matters there is unrestricted behavior and this design does not contain any. The current microphone setup also raises privacy and ambient-noise issues that were not tested.

Labels themselves carry residual uncertainty. Class membership rests on a button press confirmed by the experimenter, and the alignment audit showed that muscle activity preceded the marker at four of every five onsets, with a median lag of -0.075 s. The 0.25-s transition guard exceeds the median error in the harmful direction and covers the 95th percentile, but 7% of trial-leading windows may still carry a partially mislabeled edge. The acquisition token for the grinding condition was a filename label chosen at recording time; the corresponding class is reported throughout as instructed grinding, and performing grinding on command is plausibly more regular and easier to detect than spontaneous behavior—a bias in the optimistic direction.

Finally, the modality comparison is limited by its input before it is limited by anything else. It does not measure what a microphone at this placement contributes, because the channel it consumed is duplicated across participants, unaligned with the electromyogram and band-limited to an envelope (Section 4.2); a working transducer might do better or worse, and this study cannot distinguish those cases. Even had the channel been sound, the comparison would remain algorithm-specific—a stronger EMG-only architecture, a different transducer position, additional muscles, or calibrated intensity features could all change it—and it would remain untested against the conditions that matter most for audio: the recordings were made in a quiet laboratory, and ambient noise, speech and clothing contact were never present. The reported computational measurements were obtained on laptop and desktop hardware rather than a wearable prototype, and the acausal filter chain means they do not describe a streaming system at all.

6 Conclusion

This five-participant proof-of-concept study evaluates whether a compact dual-branch wavelet CNN can combine surface EMG and audio to classify quiet rest and four instructed awake jaw activities. Under leakage-free nested leave-one-subject-out evaluation on corrected signals, the fused model achieved a participant-level accuracy of 82.1%, a macro-F1 of 73.6%, and a macro one-vs-rest AUC of 0.962, with per-participant accuracy ranging from 72.0% to 91.7%. Excluding chewing yielded 79.8% accuracy and 77.9% macro-F1 across rest, movement, clenching, and grinding. The collapsed clenching/grinding-versus-rest/movement/chewing analysis achieved an F1 of 87.3% and an AUC of 0.966, with 6.0% of quiet-rest windows falling on the tooth-contact side. A modality ablation showed that EMG carries essentially all of this performance: audio alone reached 42.8% macro-F1 against 72.0% for EMG alone, and adding audio to EMG changed macro-F1 by less than the spread between training seeds. A subsequent audit found that the microphone channel of this collection is not per-participant audio—83 of 100 recordings carry another participant's waveform, rotated—so those figures bound what such a channel contributes and do not measure a microphone. RQ2 is therefore recorded as unanswered rather than answered weakly, and the case for carrying a microphone in a future system is untested rather than supported. The baseline architecture comparison is prespecified but not yet run, and is reported as pending rather than estimated from the superseded pipeline.

A second conclusion is methodological, and this study supplies two instances of it. The dominant factor in the electromyographic results was not the model but the measurement chain: correcting a filter that removed an absent mains fundamental while passing the harmonics that carried 91.3–99.8% of the in-band power changed the pipeline's screening macro-F1 by more than every other intervention we evaluated combined, and eliminated cases in which one participant's resting EMG exceeded another's active EMG. The second instance is the microphone channel, and it is instructive precisely because the first audit did not catch it: that audit was thorough, it was reproducible, and it was pointed at one modality. We therefore recommend two checks, both of which cost one pass over the data. Report a measured interference statistic beside every accuracy, and plot filter responses over the measured spectrum of the recordings rather than alone. And fingerprint every channel of every recording—a hash of its sorted samples is enough, and is invariant to the circular rotation that hid the defect here—then group the fingerprints and confirm that no held-out participant's signal appears in its own training set. Defects of this second kind are invisible in any single file: every affected recording here is individually well-formed and plausible, and the duplication is obvious only when two files are compared.

The study does not validate a bruxism diagnostic device. Although quiet-rest windows are now included, the study did not include bracing or thrusting, unrestricted background behavior, spontaneous awake episodes, sleep recordings, or an independent clinical reference standard. The defensible conclusion is therefore technical: surface EMG with modality-specific wavelet decomposition merits further evaluation for distinguishing instructed jaw activities from quiet rest in a larger, phenotyped, multi-day continuous-recording study designed around current bruxism terminology and clinical assessment frameworks, and a near-joint microphone merits inclusion in that study as a complementary channel—one whose measurable value here lay in separating quiet rest from tooth contact rather than in raising overall classification.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available because of participant-privacy restrictions but are available from the corresponding author on reasonable request, subject to institutional and ethical requirements.

Code availability

The custom code used for signal preprocessing, wavelet decomposition, model training, and evaluation is available from the corresponding author on reasonable request. Every number and figure in this manuscript was regenerated from a saved prediction ledger in which each held-out window appears exactly once per task, model, modality and seed, together with its full probability vector, the source commit, and hashes of the resolved configuration, the data manifest, the window index and the model checkpoint. Two runs are reported. The primary five-class run is `five_class_nested_loso_20260807T211827_2b6fb5ac`, carrying configuration hash `2b6fb5ac`; the modality ablation is `modality_and_no_chewing_20260810T020642_cead62e4`, carrying configuration hash `cead62e4`. Both share manifest hash `7ebbcc8d` and window-index hash `b2cf690c`, so every condition in this paper was evaluated on the identical set of windows. The manifest schema was subsequently extended with the microphone-integrity columns and the cross-recording fingerprint pass described in Section 4.2, which changes the manifest hash; the two run bundles retain the hash they were produced under, and their configurations now carry a written acknowledgement of the microphone defect, which the pipeline requires before a run may read that channel. Acknowledgements are excluded from the configuration hash—they record who authorised a run, not what it computes—so both configuration hashes above are unchanged and both bundles still reproduce.

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Author contributions

M.H.F.: methodology, software, analysis, writing—original draft, and writing—review and editing. A.R.: protocol development, software, data curation, data collection, formal analysis, and review and editing. R.S.: methodology, validation, and writing—review and editing. M.P.F.: conceptualization and study design, research-grant development, protocol development, and manuscript revision. R.A.: conceptualization, project administration, resources, supervision, and writing—review and editing. T.A.W.: conceptualization, research-grant authorship, protocol development, project oversight, and manuscript revision. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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